What Does Swiss German Cost a Language Model? Dialect versus Chat-Style Spelling in LLM Normalization

Erik Avtandilyan and Simone Martinecz and Kelvin Myers
University of Zurich
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https://github.com/Eroavt/gsw-eval

Abstract

Swiss German is spoken by about five million people but has no standard spelling, so every writer spells it their own way. Large language models (LLMs) understand it worse than Standard German and need more tokens for it. We ask how much of this cost comes from the dialect and how much from chat-style spelling. We build GSW-EVAL, an open, reproducible benchmark that feeds the same 500 sentences to a model as Standard German, as careful Swiss German in eight dialects, and with simulated chat spelling and character noise. On SwissDial, the step to careful Swiss German costs the 8B models 20 to 24 chrF points and Claude Sonnet 5.5 only 8.6, while simulated chat spelling adds 1 to 5 points. That simulation is mild, though: it changes only 6.5% of the words, and damaging half of all words costs Claude 9.2 points, as much as the dialect. The Swiss-focused Apertus scores 6.4 points above Llama 3.1 at equal size, and the three times larger Gemma 3 27B only 1.1 above Apertus; Claude is 11.5 points ahead of Apertus on dialect input, while the two tie on Standard German. A small dialect cost can also mislead, because Gemma rewrites even Standard German. Open LLM tokenizers need 16 to 30% more tokens for Swiss German. **[TODO: one sentence each on the fresh set and the ratings.]**

1 Introduction

Written Swiss German is everywhere in Switzerland, in messages, social media and comments. It is a low-resource variety inside a high-resource language: Standard German is one of the best-served languages in NLP, its dialects are not (Joshi et al., 2020). Two things make Swiss German hard for a language model. First, the *dialect*: the eight major dialects differ in vocabulary, sound and grammar. Second, the *spelling*: there is no orthography, so the same writer may spell a word differently from one message to the next.

Users pay for both, in quality and in money, since API prices are per token and non-standard text splits into more tokens (Petrov et al., 2023; Ahia et al., 2023). But the two causes need different fixes. A dialect problem calls for more dialect data; a spelling problem could be solved with normalization or with robust training. Existing work mostly measures one or the other. We measure both on the same sentences.

Contributions.

1. GSW-EVAL, an open benchmark that splits the quality an LLM loses on written Swiss German into a *dialect cost* and a *spelling cost*, with confidence intervals.
2. A comparison of four models: a Swiss-focused 8B model scores about 6 chrF points above a model of the same size and comes within 1.1 points of one three times its size, while a frontier model loses less than half as much to the dialect.
3. A token-tax analysis of nine tokenizers, showing that simulated chat spelling does not explain the extra tokens.
4. A small fresh test set in which each native writer translates the same sentences twice, carefully and in texting style, from five native speakers.

2 Related Work

[TODO: Kelvin: rewrite this section in your own words, 0.5 column.]

Swiss German NLP. Earlier work normalizes Swiss German transcripts to a common spelling with character-level machine translation (Scherrer and Ljubešić, 2016) on the ArchiMob corpus (Samardžić et al., 2016), but that is transcribed speech, not text people write themselves. The Swiss SMS corpus (Stark et al., 2014) holds real

messages, but offers no parallel versions across dialects, so it cannot support a controlled comparison. SwissBERT (Vamvas et al., 2023) adapts an encoder to Swiss national languages, and Vamvas et al. (2024) adapt encoders to written Swiss German. SwissDial (Dogan-Schönberger et al., 2021) provides the same sentences in eight dialects with Standard German references, which makes a controlled comparison across dialects possible, at the price of one writer per dialect. Collecting dialect data remains the bottleneck (Sun et al., 2025).

Dialect and noise robustness. Aepli et al. (2023) show that standard MT metrics are unreliable when the output is Swiss German. Our outputs are Standard German, where chrF is on firmer ground, but we still check it against human ratings. DialectBench (Faisal et al., 2024) shows large gaps between standard and non-standard varieties across many tasks, but does not ask how much of a gap is spelling. Synthetic character noise breaks neural translation (Belinkov and Bisk, 2018), and injecting such noise during training helps transfer between closely related languages (Aepli and Sennrich, 2022). We use the same noise for evaluation, and vary its strength, because a single noise level says little about robustness.

Tokenization cost. The same content needs very different numbers of tokens in different languages (Rust et al., 2021; Petrov et al., 2023), which makes commercial models more expensive for some users (Ahia et al., 2023). We apply this lens *within* one language: Standard German versus its dialects.

3 Data

SwissDial. SwissDial v1.1 (Dogan-Schönberger et al., 2021) contains 11,213 Standard German sentences with transcriptions in up to eight dialects (AG, BE, BS, GR, LU, SG, VS, ZH), one writer per dialect. The versions are free translations, so they share their meaning but not their wording. We keep the 2,529 sentences available in all eight dialects and drop 232 marked as code-switching, leaving 2,297. We draw a fixed split of 50 development and 500 test sentences (seed 13), giving 4,000 Swiss German test inputs per condition. The Standard German sentence is always the reference.

Swiss German NTREX. To check that our findings are not specific to SwissDial, we repeat the evaluation on a second, independent test set. Aepli et al. (2023) had two professional translators translate the 1,997 English news sentences of NTREX-128 (Federmann et al., 2022) into Bernese and Zurich German; NTREX-128 itself provides a Standard German translation of the same sentences, which serves as the reference. 1,850 sentences can be aligned in both dialects, and we fix 500 of them as a second test split. Both the Swiss and the Standard German versions are translated from English, so the reference is looser and absolute scores are lower.

Fresh set. Five native speakers from **[TODO: cantons]** each translated 24 Standard German sentences (20 everyday chat, 4 news-like) twice on their phone: first the way they would text a friend, then, in a separate round, carefully. The chat round comes first so it stays natural, and nothing is prefilled, so neither version copies the other. Because the same person writes both versions, the difference between them is spelling and style, not dialect. The set was written after the models were trained, so it cannot be in their training data. It is released with the code (CC BY 4.0).

4 Method

Conditions. Every test sentence is given to each model in four forms: (1) *Standard German input*: the reference itself; (2) *clean*: the SwissDial transcription; (3) *texting*: the clean text rewritten by 23 texting rules (lower case, dropped punctuation, collapsed double vowels, short forms such as *vlt* for *vilicht*), each firing with a probability, **[TODO: Simone: confirm the rules are realistic]**; and (4) *character noise*: 10% of words get one random letter insertion, deletion or substitution, following Aepli and Sennrich (2022). All noise is seeded per sentence. The texting rules are mild: ignoring case and punctuation they change 6.5% of words, and 57% of inputs not at all. A stress test therefore raises the noise to 25, 50 and 100% of words on 100 sentences.

Baselines. The *identity* baseline copies the Swiss German input unchanged; it is the lower bound. Standard German input is the upper bound.

Dialect and spelling cost. We define the *dialect cost* as chrF on Standard German input minus chrF on clean Swiss German, and the *spelling cost* as chrF on clean minus chrF on texting input. Both use the same source sentences, so topic and meaning are held constant, not wording. The dialect cost covers all that separates careful Swiss German from Standard German: vocabulary, grammar and each writer's own spelling. The split is thus between

the careful written dialect and chat-style changes on top of it, not between dialect and spelling in general.

Models. Apertus 8B Instruct (Hernández-Cano et al., 2025), trained with a strong Swiss and multilingual focus; Llama 3.1 8B Instruct (Grattafiori et al., 2024), the same size without a Swiss focus; and Gemma 3 27B Instruct (Gemma Team, 2025), a larger open model without a Swiss focus. All three run locally with 8-bit weights and greedy decoding via MLX.[1] Claude Sonnet 5.5 runs through its API with the lowest thinking setting, as a commercial reference. All models receive the same German prompt asking for the Standard German sentence only. It calls the input Swiss German even when it is Standard German, which may invite rewriting.

Metric and statistics. We score with chrF (Popović, 2015) as implemented in sacreBLEU (Post, 2018) (`nrefs:2|case:mixed|eff:yes|nc:6|nw:0|space:no|version:2.6.0`); character n-grams suit a task where most errors are partial word forms. All intervals are 95% bootstrap intervals (Koehn, 2004) that resample whole sentences with all their dialects, because the eight versions of a sentence are not independent. Model differences use the same bootstrap, paired, with 10,000 resamples. SwissDial writes numbers as words (*zweitausendelf*) while models mostly write digits (*2011*); we therefore score against both versions of the reference, which SwissDial provides, and keep the better match (multi-reference chrF). We also remove answer preambles such as "Der Satz lautet auf Standarddeutsch:".

Token tax. We compare nine tokenizers, those of our models and of Qwen 3 (Yang et al., 2025), Mistral (Jiang et al., 2023), OpenAI's o200k, XLM-R (Conneau et al., 2020) and SwissBERT. For a tokenizer, the token tax is the number of tokens of the Swiss German sentences divided by the tokens of the same sentences in Standard German, summed over all 2,297 parallel sentences and eight dialects. Because the texting rules drop punctuation and punctuation marks are tokens, we compare all conditions with punctuation removed.

Human evaluation. The same five people rated 112 outputs for adequacy on a 1–5 scale: all four models on 24 inputs, plus 16 model rewrites of Standard German input, which show how much of the gap to 100 is harmless paraphrase. Each output was rated by two of them, blind to the model and without a reference, so they judged meaning, not word overlap, and for scores of 3 or lower they chose what went wrong. Rater pairs vary, so we report Krippendorff's α (Krippendorff, 2011). **[TODO: α, Spearman chrF vs rating, and whether chrF picks the same better model.]**

[1] https://github.com/ml-explore/mlx-lm

5 Results

Careful dialect costs far more than simulated chat spelling. Table 1 shows that moving from Standard German to clean Swiss German costs Apertus 19.9 and Llama 23.8 chrF points, but Claude only 8.6. Texting spelling costs a further 3.5, 5.0 and 1.1 points, character noise 2.9, 3.0 and 1.4. All models stay far above the identity baseline (46.9), so they translate rather than copy. Much of the texting loss is case and punctuation, which chrF counts: the identity baseline loses 6.8 points to texting, but only 0.6 once both are ignored.

Stronger spelling damage does cost quality. Table 2 raises the noise for Claude. The loss grows faster than the noise: 1.6 points at 10% of words, 4.2 at 25%, 9.2 at 50%, where it matches the whole dialect cost, and 28.5 when every word is damaged. The small spelling cost is thus a result of a small dose, not proof that spelling does not matter. Random noise is not real texting, so the fresh set has to show where real writers fall on this curve. **[TODO: local models from the MacBook Pro run, and the dose of the fresh set.]**

The Swiss-focused model leads at 8B; Claude leads overall. Apertus beats Llama in every condition (paired bootstrap, $p < 0.01$): by 2.5 points on Standard German input, 6.4 on clean dialect and 7.9 on texting input. This fits the idea that Swiss text helps, but does not prove it: the two models differ in more than their data, the gap falls to 2.5 under one prompt wording, and on NTREX it is 2.9 and does not grow on texting input relative to Standard German (3.4). Gemma, three times larger, is only 1.1 points ahead of Apertus on clean dialect and 2.0 on texting input. Claude is 11.5 points ahead of Apertus on clean dialect and 13.9 on texting input, but the two tie on Standard German input ($+0.2$, interval -1.1 to $+1.5$), so Claude's lead appears only on dialect input (Table 3).

Even Standard German is not copied. No model reaches 100 on Standard German input: Apertus returns the sentence unchanged 64% of the time, Claude 59%, Llama 45% and Gemma only 23%; the rest are paraphrases such as "Jemand

Model	Std. German	Clean	Texting	Char. noise	Dialect cost	Spelling cost
Identity (lower bound)	100.0 ±0.0	46.9 ±1.0	40.0 ±0.8	45.3 ±1.0	53.1 ±1.0	6.8 ±0.4
Apertus 8B	94.0 ±1.0	74.1 ±1.2	70.6 ±1.2	71.3 ±1.1	19.9 ±1.2	3.5 ±0.5
Llama 3.1 8B	91.5 ±1.0	67.7 ±1.2	62.7 ±1.3	64.8 ±1.2	23.8 ±1.2	5.0 ±0.6
Gemma 3 27B	84.7 ±1.1	75.2 ±1.1	72.6 ±1.1	73.3 ±1.1	9.4 ±1.2	2.6 ±0.5
Claude Sonnet 5.5	94.2 ±1.0	85.6 ±0.9	84.5 ±0.9	84.2 ±0.9	8.6 ±1.1	1.1 ±0.3

Table 1: chrF on the 500 test sentences (±: half-width of the 95% bootstrap interval). Swiss German columns pool 4,000 inputs over eight dialects. Dialect cost = Standard German input minus clean; spelling cost = clean minus texting.

Input	Words changed	Apertus	Llama	Claude
Simulated chat spelling	6%	3.9	5.3	1.6
Noise in 10% of words	9%	–	–	1.6
Noise in 25% of words	25%	–	–	4.2
Noise in 50% of words	49%	–	–	9.2
Noise in 100% of words	98%	–	–	28.5

Table 2: Stress test on the first 100 test sentences: chrF lost compared with careful spelling. Words changed ignores case and punctuation.

Bernese input: *Si het me drvo gno.*		
Reference	Sie hat mehr davon genommen.	
Apertus	Sie hat mir davon erzählt.	39
Llama	Ich kenne dich.	8
Gemma	Sie hat mich dafür bestraft.	25
Claude	Sie hat mehr davon genommen.	100

Table 3: The sentence with the largest gap between models, with sentence chrF. Short dialect function words such as *me* (more) and *drvo* (of it) mislead the open models.

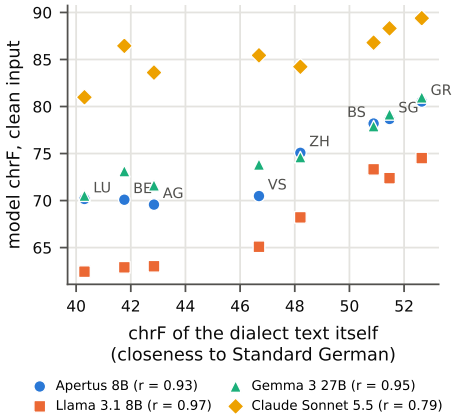


Figure 1: Per-dialect chrF of each model (clean input) against the chrF of the untranslated dialect text, a measure of how close its spelling is to Standard German.

vermiest mir hier die Arbeit" for "Irgendjemand pfuscht mir hier ins Handwerk". This is why the upper bound matters: part of every model's gap to 100 is paraphrase that chrF counts as error, not misunderstanding. It also means a small dialect cost alone is not good news. Gemma's dialect cost (9.4) is almost as small as Claude's (8.6), but mainly because Gemma rewrites even Standard German (84.7 against Claude's 94.2), not because it reads dialect well. The upper bound also depends on the prompt: Llama falls from 91.7 to 85.1 under one wording. The dialect cost has to be read together with the upper bound. **[TODO: the human ratings tell how big the paraphrase part is.]**

Hard dialects are the ones written farthest from Standard German. Scores range from 69.6 (AG) to 80.5 (GR) for Apertus, and the ranking is almost the same for Llama and Gemma (Figure 1). All models correlate strongly with the identity score of each dialect ($r = 0.93$, 0.97 and 0.95 for the open models, 0.79 for Claude, range 81.0 to 89.4): models do worst where the written form is farthest from Standard German. With only eight dialects these estimates are rough (Claude's 95% interval is about 0.2 to 0.96), so we do not claim that Claude depends on it less. Since SwissDial has one writer per dialect, this mixes the dialect with the writer's personal spelling. The fresh set separates the two. **[TODO: fresh-set result.]**

Simulated chat spelling does not explain the token tax. Open LLM tokenizers need 16% (Mistral) to 30% (Apertus) more tokens for Swiss German than for the same content in Standard German, and the encoder SwissBERT 59% (Table 4). Texting spelling changes this by 1.3% at most for these LLM tokenizers. Bernese is the most expensive dialect and Zurich German the cheapest for all LLM tokenizers. Claude's tokenizer behaves differently: counted on the test split with Anthropic's endpoint, it needs about 3 tokens per word for both, so Swiss German costs no extra there, and texting spelling

Tokenizer	Tok./word	Tax	Range	Texting
Apertus 8B	1.64	1.30	1.23–1.40	+0.3%
Llama 3.1 8B	1.93	1.17	1.12–1.25	-1.3%
Gemma 3	1.62	1.25	1.20–1.35	+0.0%
Qwen 3	1.92	1.18	1.12–1.26	-1.0%
Mistral 7B v0.3	2.10	1.16	1.11–1.23	-0.7%
OpenAI o200k	1.60	1.27	1.23–1.37	-0.4%
XLM-R	1.63	1.30	1.25–1.39	+0.8%
SwissBERT	1.43	1.59	1.48–1.70	+7.1%

Table 4: Token tax per tokenizer. Tok./word: tokens per word of Standard German. Tax: tokens for Swiss German divided by tokens for the same Standard German sentences (punctuation removed, eight dialects pooled). Range: cheapest to most expensive dialect. Texting: change of the tax caused by texting spelling.

even saves 9%.

A second test set shows the same order. On Swiss German NTREX, clean input gives Claude 59.9, Gemma 57.5, Apertus 56.6 and Llama 53.8 (neighbouring models differ with $p < 0.01$), and texting costs 0.4 to 2.3 points. We do not interpret its dialect cost: the Swiss and Standard German versions are separate translations from English, so part of the gap is wording, not dialect.

Robustness. Across three prompt wordings (100 test sentences), clean-dialect chrF moves by 0.2 points for Claude, 2.2 for Llama and 3.1 for Apertus. Apertus stays ahead of Llama under every wording, and the spelling cost stays between 3.1 and 5.3 points for both 8B models, far below their dialect cost, which itself moves between 16.5 and 21.7. Full-precision (bf16) weights change the 8-bit scores of Apertus and Llama by at most 0.5 points.

Human ratings. **[TODO: kappa, Spearman chrF vs rating, and whether chrF picks the same better model as the raters.]**

Fresh set. **[TODO: real vs synthetic texting cost, per writer.]**

6 Discussion and Limitations

[TODO: Kelvin + Erik: write after all results are in. Points to cover:] (1) SwissDial transcriptions are free translations by one writer per dialect, so dialect and writer are confounded; the fresh set addresses this only for **[TODO: N]** writers. (2) The texting rules are mild and hand-written; real texting is richer (abbreviations, emoji, code-switching). The stress test shows that the spelling result depends on this dose. (3) chrF punishes valid paraphrases; the human ratings estimate how much. (4) We test one prompt per model in the main run and 8-bit weights; the robustness checks cover both on a subset. (5) SwissDial has been public since 2021 and may be in the models' training data; the fresh set cannot be. (6) A small dialect cost can come from a weak upper bound: a model that paraphrases Standard German (Gemma) looks robust to dialect. The cost must be read with the upper bound and copy rate. (7) Comparing models is not an ablation: we cannot attribute Apertus's lead to its Swiss data.

7 Conclusion

What does Swiss German cost a language model? Mostly the step from Standard German to the careful written dialect. It costs 8B models 20 to 24 chrF points, and open LLM tokenizers 16 to 30% more tokens. Simulated chat spelling adds only 1 to 5 points, but it is a mild test, and stronger spelling damage costs more. A Swiss-focused 8B model nearly matches a model three times its size, and a frontier model cuts the dialect cost to 8.6. The dialect cost should always be read with the upper bound. **[TODO: one sentence on the fresh set and the ratings.]**

Code and Data

Code, outputs of every model, and the fresh set: `https://github.com/Eroavt/gsw-eval`. Every number in this report can be reproduced with one command, `scripts/reproduce.sh`. It needs no GPU and no API key, because all model outputs are cached, only the SwissDial text from ETH Zurich. It ran unchanged on a fresh clone.

Use of AI Assistants

An AI coding assistant (Claude Code) wrote most of the code, the website and the voting tool, drafted this report, and wrote the first version of the texting rules. Our proposal set the research question and the design. **[TODO: Team, before submission: make this statement true and complete. Say who checked which numbers, which text is your own, and what native speakers did. Adapt it to the UZH rules.]**

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